Blood Collection, Processing and Storage 10

10.1 Overview

Approximately 20 ml of blood will be collected from all participants in the Ghana Breast Health Study, both cases and controls, during the clinic visit. Participants will first be screened to ensure that it is appropriate and safe to draw blood from them. The blood draw will consist of one 10 ml red-top (serum) tube and one 10 ml purple top (EDTA) tube.

The blood tubes will be processed and aliquotted as five different blood specimen types: serum, blood clot, plasma, buffy coat, and red blood cells. The processed specimens will be stored in freezer boxes in the -80°C freezer until they are shipped to the NCI Biorepository.

10.2 Blood Kits and Labels

Inside the biospecimen Master Kit there is a Blood Collection Kit with a red circle label on it and a Blood Processing Kit with a green circle label on it. Also in the Master Kit is a pink Blood Form. Verify the root Sample ID (GBR#####) on the Blood Collection Kit bag (GBR#####-BK01), red-top tube (GBR#####-RT01), purple top tube (GBR#####-LT01), Blood Processing Kit (GBR#####-PK01), and cryovials (GBR##### -0011, -0012, -0013, -0021, -0031, -0032, -0041, -0042, -0043, -0051) match the Sample ID on the Master Kit bag. As noted in Chapter 7, if there are mis-matched Sample IDs within the Master Kit, the entire Master Kit and forms within it should be discarded and a new Master Kit and Sample ID should be assigned to the participant.

The Blood Collection Kit contains the following necessary items to collect blood specimens from the participant:

- One plastic 10-ml red-top vacutainer tube;
- One plastic 10-ml purple-top vacutainer tube;
- 2 examination gloves;
- Tourniquet;

- 2 Alcohol wipes (two per draw);
- 2 - 2" x 2" gauze pads (four per draw);
- 21g Butterfly needle blood collection set with 12" tubing and adapter; and
- Band-Aid,

Additional supplies needed or available for blood draw, but not included in specimen collection kit:

- Transpore tape;
- Tube rack;
- Chux Pad; and
- Sharps container.

10.3 Completing the Blood Form and Pre-Collection Questions

The Participant ID (PID) label should already have been placed on the pink Blood Form (Appendix 10-1) in preparation for collecting biospecimens. If the PID label is missing, the Interviewer should place it on the form in the upper left hand corner. The Interviewer should also verify that the root Sample ID (GBR#####) on the Blood Form (GBR#####-BK01) matches the Sample ID on all the Blood Collection Tubes (GBR#####-RT01, -LT01).

The Interviewer should confirm that the participant has provided her consent to participate in the study and is willing to provide a blood sample. If the participant did not initial on the Informed Consent that she agrees to provide a blood sample, then do not collect the sample. Regardless of whether or not a blood sample is collected, the Interviewer will need to complete the appropriate questions contained in the shaded Collection Summary Box.

In the Collection Summary Box, the Interviewer should circle the corresponding hospital listed in item A. After confirming whether the participant consented to provide blood, the Interviewer should circle 1 for "YES" if she <u>did initial</u> on the Informed Consent that she agreed to provide a blood sample or 2 for "NO" if she <u>did NOT initial</u> on the Informed Consent that she agreed to provide a blood sample. If the participant did not agree to provide a blood sample, do not make arrangements with the phlebotomist and promptly return the Blood Collection and Processing Form to the participant file for data entry.

If the participant has agreed to provide a blood sample, then proceed to Questions 1 through 3 in Section 1 on page 2. Ask the participant Questions 1, 2, and 3 exactly as written and record the answers. Participants who have hemophilia, who are currently taking blood thinning medications, and who received chemotherapy or blood products in the past four weeks should not provide blood for the study. If any of the questions in Section 1 have a "YES" answer, **then do not request the phlebotomist to collect blood from the participant** and record "NO" for item C in the Collection Summary box. If all answers to questions in Section 1 are a "NO", then return back to the front and record "YES" for item C. If the participant is able to provide a blood sample, then proceed with making arrangements for the phlebotomist to draw the blood.

Before blood is drawn from the participant, the phlebotomist should confirm the answers circled under items A, B, and C in the Collection Summary Box are correct. The participant should be asked Questions 1 – 3 on page 2 again, by the phlebotomist to confirm the participant is able to provide blood and if it is safe to do so. If the participant responds "YES" to any of the questions in Section 1, **then do not collect blood from the participant**. The phlebotomist should cross-out the "NO" that was circled and circle "YES". Next to the change, the phlebotomist should write his/her initials authorizing the change. The phlebotomist must contact the Interviewer immediately to notify them of the change and return the Blood Form to the participant's file.

If all answers to questions in Section 1 are confirmed as "No", then the phlebotomist should continue to ask questions 4 - 14 in Section 2 of the Blood Form. After all of the participant's answers have been recorded on the form, the phlebotomist can proceed with the blood draw.

In some rare situations, a participant agrees and is able to provide a blood sample, but blood is never drawn for various reasons, such as the phlebotomist not being available or the participant not feeling well and returning home. In these cases, the Interviewer should complete items A,B, and C as previously described. Additionally, the Interviewer should circle 3 for "Not Collected" for both the Red Top Tube (Item E) and the Purple Top Tube (Item F) and provide a comment next to items E and F as to why the blood was never collected. Item D should be left blank. The Blood Form should be promptly returned to the participant's file.

Instructions for completing the form can also be found in Appendix 10-2.

10.4 Blood Draw Procedures

The phlebotomist will record the date and time of the blood draw (Item D) in the shaded Collection Summary Box on the Blood Form. Following the blood draw, the phlebotomist will record whether each tube was full, partial, or not completed (Items E and F). If a participant is reluctant to provide blood for both tubes, or a difficult draw prevents obtaining blood for both tubes, priority should be placed on obtaining blood for the red top tube before the purple top tube.

The phlebotomist will use the following procedures to obtain blood specimens for the study:

- Follow universal precautions when drawing or handling blood to minimize the risk of transmission of infectious agents.
- The participant should be in a seated or reclined position. Clothing should not restrict access to the arm.
- Wash hands thoroughly and put on gloves and lab coat.
- Have the participant extend the arm, palm up, and straight at the elbow.
- Position the arm on the work area so that the veins are readily accessible and you are able to work in a comfortable position. Be sure that the arm is in a downward position, with the elbow lower than the heart to prevent backflow. If necessary, place a prop such as a rolled up Chux under the elbow to achieve this position.
- Inspect the arm to be used for venipuncture. The veins of choice are those located in the antecubital area. Do not draw blood from an arm that has a rash, open sore, is swollen, or shows signs of a recent venipuncture or hematoma. If the chosen arm is not appropriate, inspect the other arm. Check carefully for scar tissue or the presence of tendons near the vein and avoid these areas.
- A maximum of two draw attempts can be made.
- Apply the tourniquet several inches above the participant's elbow and probe for a suitable vein.
- Select a vein that is palpable and well-fixed to surrounding tissue. Palpate even when the vein can be seen to determine the size, depth, and direction of the vein.

If the veins do not distend rather quickly, the following techniques may be used:

- Have the participant open and close her hand several times.
- Massage the arm from wrist to elbow; this forces blood into the veins.

- Apply a warm compress for a few minutes.
- Tap the area sharply with the index and second finger two or three times; this causes the veins to dilate.
- The arm to be used for venipuncture may be hung at the participant's side without a tourniquet. This will allow the veins to fill with blood to their capacity.
- Examine the participant's other arm. Sometimes the veins in one arm are larger than in the other.
- If the blood draw can be delayed for approximately one hour, up to 7 ounces of water may be given to the participant.
- If the tourniquet has been applied for more than one minute while you search for a vein, release the tourniquet for two to three minutes before the venipuncture. Prolonged obstruction of blood flow by the tourniquet is unnecessary and uncomfortable for the participant and may affect results.
- Ask the participant to make a fist, tell her to avoid pumping or straining as this can affect the results.

Obtain the blood specimen as follows:

- Cleanse the area with an alcohol wipe. Hold the alcohol wipe with two fingers on one side, so that only the other side of the wipe touches the area of the puncture site. Cleanse the area using a circular motion beginning with a narrow radius and moving outward so as not to cross over the area already cleansed. Repeat with a second alcohol wipe. Air dry (~30-60 seconds). The area should be completely dry before the venipuncture is done in order to reduce the burning sensation caused by alcohol penetrating the skin.
- Remove the sheath from the needle or the butterfly.
  - The vein should be "fixed" or held taut during the puncture. Place the left thumb about one to two inches below the point of entry and pull the skin gently in a downward motion. (This stretches the skin and "anchors" or "fixes" the vein.) Do not fix the vein with a finger above the venipuncture site, as this puts the phlebotomist at risk for a hazardous needle stick.
  - Hold the needle in line with the vein, with the bevel up and at a 15° angle with the skin, about ½ an inch below the proposed point of entry to the vein.
  - Pinching the butterfly "wings" together, push the needle firmly and deliberately into the vein. When you are firmly in the vein, blood will appear in the tubing of the butterfly. If blood does not appear in flash window, the needle can be adjusted slightly, but do not move side to side or allow the needle to come

completely out of the skin. Carefully tape the "wings" or the tubing to the participant's arm or hand with the piece of transpore tape.

- Quickly push the red-top vacutainer onto the butterfly needle in the holder, puncturing the stopper of the vacutainer. The tube must be punctured in the center of the stopper.

- Avoid releasing the tourniquet immediately after the flow begins as this may cause the vein to collapse and the blood flow to stop. Rather, loosen it slightly but do not release it completely until the last tube is almost full. The participant may open her fist.

- Do not rock or change position of the tube while it is filling.

- Fill the tube to capacity, remove from the holder and place it in a holding rack. The order of the tubes is as follows:

  1) Red top 10 ml tube

  2) Purple top 10 ml tube

- If a bruise begins to develop during the blood draw, discontinue the procedure and immediately apply moderate pressure to the site. If the participant gives verbal consent, a second attempt may be made in the other arm. However, no more than two attempts should be made.

- Upon removing the purple-top tube from the holder, gently invert it eight times. Inversion of each tube should occur as soon as possible. Do not shake tubes. Place each tube into the holding rack.

- After the purple-top tube is filled, carefully remove the tourniquet and tape and quickly withdraw the butterfly, covering the puncture site with a sterile gauze pad. Have the participant hold the gauze pad with mild pressure and raise the arm in the air for two minutes. Do not let the participant bend the arm. This procedure will help prevent the formation of a hematoma.

- Immediately place the used needle assembly into the sharps container.

- Check the venipuncture site. If it is adequately clotted, apply tape over the gauze or a Band-Aid. Instruct the participant to remove it in no less than 45 minutes if the bleeding has stopped. Encourage the participant to avoid heavy lifting for a few hours following blood draw, particularly for those participants on anti-platelet medications. Also, suggest that the participant sit quietly for a few minutes. Observe the participant during this time for adverse effects. If bleeding continues, keep direct pressure on the site for five minutes or more.

- Discard the gloves in a biohazard waste container and wash your hands thoroughly with soap and water or hand sanitizer.

10.5 Complications of Blood Draw

10.5.1 Hematomas

Hematomas are the most common complication of venipuncture. They are masses produced by coagulation of extravasated blood in a tissue or cavity. They may result from through-and-though puncture to the vein or from incorrect insertion of the needle into the lumen of the vein, allowing the blood to leak into the tissue by way of the bevel of the needle. In the latter case, correction may be made by advancing the needle into the vein. At first sign of uncontrolled bleeding, the tourniquet should be released and the needle withdrawn. Mild pressure to the puncture site should be applied immediately. Hematomas also result from the application of the tourniquet after an unsuccessful attempt has been made to draw blood.

Hematomas most frequently result from insufficient time spent in applying pressure, from failure to apply pressure, and from the bad habit of flexing the arm to stop bleeding. Once the venipuncture is complete, the participant should be instructed to apply mild pressure to the puncture site and raise her arm straight in the air for about two minutes. Constant pressure should always be maintained until the bleeding stops. Pressure should be applied with dry, sterile gauze; do not use a wet sponge as it encourages bleeding. Band-Aids do not take the place of pressure and, if used, are not to be applied until after the bleeding stops.

Arms covered with ecchymoses (escape of blood into the tissues, producing a large and blotchy area of superficial discoloration; bruises) demonstrate poor technique or a haphazard manner. Proper techniques must be employed at all times to prevent unnecessary hematomas.

10.5.2 Red Cross Procedures for Handling Vasovagal Reaction[1]

"Fainting" is a partial or complete loss of consciousness. This is due to a reduced supply of blood to the brain for a short time. Occasionally, a person collapses without warning. Recovery of consciousness usually occurs when the victim falls. Injury may occur from the fall. To prevent fainting, a person who feels weak or dizzy should lie down or bend over with her head at the level of her knees. Signs and symptoms include:

[1] From *Standard First Aid and Personal Safety, Second Edition*, the American Red Cross, New York: Doubleday and Company, Inc., 1981, pp. 173-174.

- Extreme paleness,
- Sweating,
- Coldness of the skin,
- Dizziness,
- Numbness and tingling of the hands and feet,
- Nausea, or
- Possible disturbance of vision.

If at any time during the venipuncture procedure the participant exhibits any of the manifestations listed above, conclude the venipuncture immediately and perform the first aid procedures.

- Place the participant in a supine position.
- Loosen any tight clothing and keep crowds away.

If the participant vomits, roll her onto her side or turn her head to the side and, if necessary, wipe out her mouth with your fingers, preferably wrapped in cloth.

- Maintain an open airway.
- Do not pour water over the participant's face because of the danger of aspiration. Instead, bathe her face gently with cool water.
- Do not give any liquid unless the participant has revived.
- Examine the participant to determine if she has suffered injury from falling.
- Seek medical assistance. Do not leave the participant unattended. Observe the participant carefully afterward because fainting might be a brief episode in the development of a serous underlying illness.

An Adverse Events Report must be completed if the participant faints, falls, or vomits during the blood draw.

10.5.3 Continued Bleeding

Some participants may be receiving certain drug therapies or have bleeding disorders that may cause them to continue to bleed after the venipuncture. To prevent bleeding, it may be necessary to apply pressure to the puncture site for an extended period of time. Apply additional pressure and observe the participant until bleeding has stopped. Apply a pressure bandage.

10.6 Transport of Blood Collection Tubes to Processing Laboratory

The phlebotomist should transport the blood tubes, Blood Form, and Blood Processing Kit to the processing laboratory as soon as possible after the blood draw. The blood tubes should be kept at room temperature. Avoid exposing the blood tubes to fluctuating temperature conditions or extreme ambient temperature conditions.

Blood specimens should be processed and frozen two to four hours after the time of draw. Allow the red top tube to clot for at least 60 minutes before centrifuging to avoid RBC contamination of the serum sample.

Prior to receipt of a freezer at PLH, special provisions were made for blood samples collected at PLH to be transferred to KATH for processing. Upon receipt of a freezer at PLH on June 6, 2014, samples collected at PLH are processed and stored at PLH by the same specifications described in this protocol.

10.7 Processing Blood Specimens

The Blood Processing Kit has a green circle label and contains pre-labeled cryovials for aliquotting the processed blood. It is important that the pre-labeled blood tubes are processed into the correct pre-labeled cryovials to ensure the correct material type is associated with the correct cryovial.

The Blood Processing Kit includes:

- Transfer pipettes

- 10 pre-labeled cryovials labeled as following:

Material Type	Sequence No.
Cryovials for Plasma	0011-0013
Cryovial for Buffy Coat	0021
Cryovial for RBC	0031-0032
Cryovials for Serum	0041-0043
Cryovial for Clot	0051

NOTE: If any of the pre-labeled items are not used, they should be discarded. Pre-labeled items should not be used for another study participant.

Additional supplies needed or available for blood processing, but not included in specimen collection kit:

- Centrifuge;
- Non-latex examination gloves;
- Tube rack;
- Chux Pad;
- 5 x 2" 81 slot freezer boxes; one for each material type
- BioHazard waste/Sharps container.

At the processing laboratory, the red-top vacutainer will be processed and separated into serum and blood clot. The purple-top vacutainer will be processed and separated into plasma, buffy coat, and red blood cells.

The table below shows the aliquots to be created from each tube and the sequence number to be assigned to each aliquot.

Ghana Breast Study Blood Processing				
	Material Type	Seq. Number	Volume	Comment
Red Top Tube	Serum	0041	1.8 mL	
	Serum	0042	1.8 mL	
	Serum	0043	1.8 mL	
	Clot	0051	15.0 mL	Will remain in Ghana
Purple Top Tube	Plasma	0011	1.8 mL	
	Plasma	0012	1.8 mL	

	Plasma	0013	1.8 mL	
	Buffy Coat	0021	0.5-1.0 mL	
	RBC	0031	1.8 mL	
	RBC	0032	1.8 mL	

The Processing Summary Box on the Blood Form must be completed by the laboratory technician after the red-top and/or purple-top tube(s) have been processed. The Blood Form must be returned to the Study Manager once completed.

10.7.1 Processing of Red Top Tube

10.7.1.1 Centrifuge Red Top Tubes

Process all blood specimens in a biosafety hood using approved sterile techniques from your institution and specimen handling precautions.

- Centrifuge. Prior to centrifuging, allow the red top tubes to clot for at least 60 minutes. Red top tubes should be centrifuged within two hours of collection to avoid red blood cell (RBC) contamination of the separated undiluted serum sample. Centrifuge each collected Red Top tube at room temperature (18-25°C) for a minimum of 10 minutes at 1500g/Relative Centrifugal Force (RCF).

  WARNING: Excessive centrifuge speed (over 2000g) may cause tube breakage and exposure to blood and possible injury.

- Opening the tube. To obtain an undiluted serum sample, carefully remove the stopper by grasping the red top tube firmly with one hand and placing the thumb under the stopper. (For added stability, place arm on solid surface). With the other hand, twist the stopper while simultaneously pushing up with the thumb of the other hand only until the tube stopper is loosened. Move thumb away before lifting stopper. Do not use thumb to push stopper off tube. Carefully remove the rubber stopper from tube. Do not reassemble the stopper and tube.

  CAUTION: If the tube contains blood, an exposure hazard exists. To help prevent injury during closure removal, it is important that the thumb used to push upward on the closure be removed from contact with the tube as soon as the stopper is loosened.

10.7.1.2 Aliquot Serum

- Aliquot serum. Before aliquotting serum, verify that the sample ID number on the red-top tube and sample ID on the pre-labeled cryovials are identical. When aliquotting be sure not to touch or disturb the blood clot with the tip of the pipette.
- Using a sterile graduated pipette (included in the Blood Processing Kit) aspirate the serum directly from the red top tube into the pre-labeled cryovials with sequence numbers 0041, 0042, and 0043.
- Create three 1.8 ml serum aliquots. If less than a full tube of blood is collected and there is not enough blood for three full 1.8 ml aliquots, please make multiple aliquots of less than 1.8 ml.

10.7.1.3 Storage for Serum

Serum specimens processed for the study will be stored in 2-inch freezer boxes in a -80° freezer. Serum specimens will be stored together. Other specimen types will be stored separately.

- Transfer serum aliquots 0041, 0042 and 0043 into a 2" box with a 9x9 grid (space for 81 vials). Only serum aliquots will be stored in this box.
- Do not store empty vials in the boxes. If one or more vials of serum were not collected, please make sure to accurately record on the Blood Collection and Processing Form which vials were not collected and then discard the empty vials.
- All vials should be frozen and stored standing up (not on their sides or with caps facing downwards).
- Mark the lid of the box with the starting and ending sample ID numbers and sequence numbers of the cryovials stored in the box, as well as its contents. Mark the lid as follows: "Serum, Sample ID GBR##### to GBR######, Sequence number 0041 to 0043".
- Store the box in a -80ºC freezer until it is ready for shipment to NCI.

10.7.1.4 Aliquot Blood Clot

- Remove the blood clot from the bottom of the red top tube and place it in the pre-labeled 15-mL cryovial labeled with sequence number 0051. Be sure the sample ID on the original red top tube matches the sample ID on the cryovial.

10.7.1.5 Storage of Blood Clot

Blood clot specimens processed for the study will be stored in 2-inch freezer boxes in a -80°C freezer. Blood clot specimens will be stored together. Other specimen types will be stored separately.

- Transfer the 15ml cryovial 0051 into a 2" box which has storage space for 9 tubes using a 3x3 grid. Only blood clots will be stored in this box. Store the box in a -80ºC freezer.
- All vials should be frozen and stored standing up (not on their sides or with caps facing downwards).
- Mark the lid of each box with the starting and ending sample ID number for each box, as well as its contents. Mark the lid as follows: "Blood Clot, Sample ID GBR##### to GBR#####, Sequence number 0051".
- Blood clot specimens will remain in Ghana. They will not be shipped to the United States.

10.7.2 Processing of Purple Top Tubes

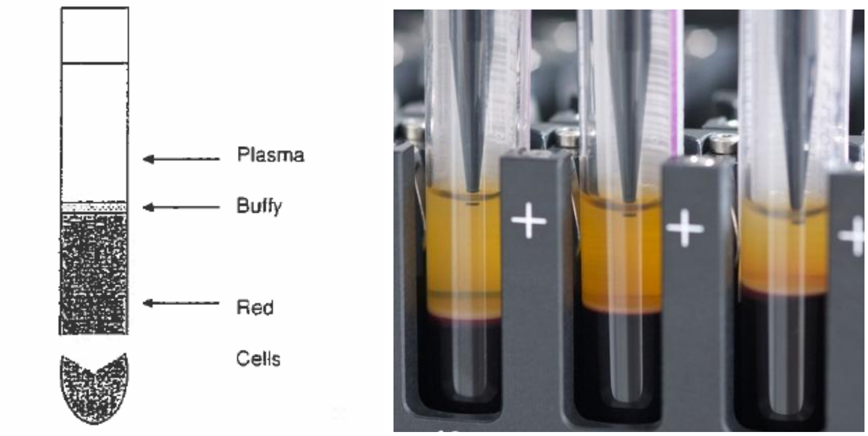
10.7.2.1 Centrifuge Purple Top Tubes

Process all blood specimens in a bio-safety hood using approved institutional guidelines, sterile techniques, and specimen handling precautions.

- Centrifuge. Purple top tubes should be centrifuged and processed within two hours of collection to avoid red blood cell (RBC) contamination of the separated plasma layer. Centrifuge each collected purple-top tube at room temperature (18-25ºC or 64-77ºF) for a minimum of 15 minutes at 1500g / Relative Centrifugal Force (RCF).

  WARNING: Excessive centrifuge speed (over 2000g) may cause tube breakage and exposure to blood and possible injury.
- For the purple top tube, simply unscrew the cap.
- After centrifuging, the blood will separate into three distinct layers as shown in the figure below: the upper layer, plasma; the lower layer, red blood cells, and a thin interface layer between the plasma and red cells that contains the buffy coat. (Buffy coat contains leukocytes and platelets and makes up approximately 1% of the entire whole blood composition).

Figure 10-1. Distinct layering of plasma, buffy coat, and red blood cells.



10.7.2.2 Aliquot Plasma

- Before aliquotting the plasma, verify that the sample ID number on the purple top vacutainer tube and sample ID on the pre-labeled cryovials are identical.
- Using a sterile pipette, slowly aspirate the plasma from the tube down to approximately 0.5 centimeters above the buffy coat interface. Be careful not to touch, mix or disturb the buffy coat (white pinkish) interface or packed red blood cells.
- Create three 1.8 ml plasma aliquots. Use the pre-labeled cryovials with sequence numbers 0011, 0012, and 0013 for these aliquots. Leave a layer of plasma in the vial so that the buffy coat layer is not disturbed by the pipetting. If less than a full tube of blood is collected and there is not enough plasma for three full 1.8 ml aliquots, please make multiple aliquots of less than 1.8 ml.

10.7.2.3 Storage for Plasma Aliquots

Plasma specimens processed for the study will be stored in 2-inch freezer boxes in a -80°C freezer. Plasma specimens will be stored together. Other specimen types will be stored separately.

- Transfer plasma aliquots 0011, 0012 and 0013 into a 2" specimen box which has storage space for 81 tubes. Only plasma aliquots will be stored in this box.
- Do not store empty vials in the boxes. If one or vials aliquots of plasma were not collected, please make sure to accurately record on the Blood Collection and Processing Form which vials were not collected and then discard the empty vials.

- All vials should be frozen and stored standing up (not on their sides or with caps facing downwards).
- Mark the lid of the box with the starting and ending sample ID numbers and BSI IDs of the cryovials that are stored in the box. Mark the lid as follows: "Plasma, Sample ID GBR###### to GBR######, Sequence number 0011 to 0013".
- Store the box in a -80ºC freezer until it is ready for shipment to NCI.

10.7.2.4 Aliquot Buffy Coat

The buffy coat is sometimes difficult to see and collect. It can be viscous and can appear as a thin milky white or pinkish colored layer just above the packed red cell layer. Some contamination of the buffy coat with the red blood cell (RBC) layer is to be expected. However, specimens that have an extremely high white blood cell (WBC) count will have a thicker and more defined buffy coat layer that will be easier to collect with minimal RBC contamination.

Before aliquotting the buffy coat, verify that the sample ID on the label matches the label on the purple top vacutainer.

- To collect, slowly aspirate the buffy coat and any remaining plasma, using a slow circular motion. Collect a total volume of 0.5-1.0 ml of buffy coat to create one aliquot.
- Pipette the buffy coat into the cryovial pre-labeled with sequence number 0021 .

10.7.2.5 Storage for Buffy Coat

Buffy coat specimens processed for the study will be stored in 2-inch freezer boxes in a -80°C freezer. Buffy coat specimens will be stored together. Other specimen types will be stored separately.

- After aliquotting has been completed, transfer the buffy coat aliquot 0021 into a 2" specimen box. Only buffy coat samples will be stored in this box. Mark the lid of the box with the starting and ending sample ID number and sequence number. Mark it as follows: "Buffy Coat, Sample ID GBR###### to GBR######, Sequence number 0021".
- Do not store empty vials in the boxes. If no buffy coat was collected, please make sure to accurately record on the Blood Collection and Processing Form which vials were not collected and then discard the empty vials.

- All vials should be frozen and stored standing up (not on their sides or with caps facing downwards).

- Store the box in a -80ºC freezer until it is ready for shipment to NCI

10.7.2.6 Aliquot Red Blood Cells (RBC)

- Before dispensing the RBCs, verify that the sample ID number on the purple top tube and the sample ID on the pre-labeled cryovials are identical.
- Using a sterile pipette, aspirate the packed red cells from the purple top tube and create two 1.8 ml aliquots. Use the cryovials labeled with sequence numbers 0031 and 0032.

10.7.2.7 Storage for Red Blood Cells

Red blood cell specimens processed for the study will be stored in 2-inch freezer boxes in a -80°C freezer. RBC specimens will be stored together. Other specimen types will be stored separately.

- After aliquotting is complete, transfer the red blood cell aliquots 0031 and 0032 into a 2" specimen box, which has storage space for 81 tubes. Only aliquots of red blood cells will be stored in this box.
- Do not store empty vials in the boxes. If one or vials aliquots of red blood cells were not collected, please make sure to accurately record on the Blood Collection and Processing Form which vials were not collected and then discard the empty vials.
- All vials should be frozen and stored standing up (not on their sides or with caps facing downwards).
- Mark the lid of the box with the starting and ending sample ID numbers and sequence numbers of the cryovials stored in the box, as well as its contents. Mark the lid as follows "RBC, sample ID GBR##### to GBR#####, Sequence number 0031 to 0032.
- Store the box in a -80ºC freezer until ready for shipment to NCI.

10.8 Menstrual Cycle Follow-Up with Participants

After the Blood Form has been returned to the study clinic, the Study Manager or Assistant Study Manager will need to review the Blood Form to determine if the participant is still having menstrual cycles (Question 13 = “Yes”). If “Yes”, the Study Manager or Assistant Study Manager will contact the participant 30 days after blood collection and ask her Question 15 in Section 3 on the Blood Form, which is the date of her next menstrual cycle following the blood draw. This should be the actual date of the cycle, not an estimate of when it is expected to occur. If the participant indicates her menstrual cycle has not started again, call back every ~14 days. The start date of the menstrual cycle following the blood draw should be recorded on the form.

10.9 Form Review

Regardless of whether or not a blood sample is collected, the appropriate questions contained in the shaded Collection Summary and Processing Summary Boxes should be completed.

- Check the form for completeness and legibility.
- Complete any items accidentally left blank.
- If any problems are encountered with the blood collection, blood processing, or completing the Blood Form, bring them to the study manager’s attention as soon as possible.